

स्वच्छंद IAS - संभव BATCH

Math: SI & CI | Reasoning: Counting of Figures

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SSC CGL Tier-II Pattern | 50 Hard Level MCQs | Pure Practice Set (No Key Included)

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PART - A: MATHEMATICS (SIMPLE & COMPOUND INTEREST)

Q1. A sum of money amounts to ₹28,224 in 2 years and to ₹29,635.20 in 3 years when invested at compound interest, compounded annually. Find the principal sum.

- (A) ₹25,000
- (B) ₹24,000
- (C) ₹22,000
- (D) ₹26,500

SSC CGL Tier-II 2022

Q2. The difference between compound interest and simple interest on a certain sum for 3 years at 15% p.a. is ₹2,835. Find the sum.

- (A) ₹35,000
- (B) ₹45,000
- (C) ₹40,000
- (D) ₹50,000

SSC CGL Tier-II 2023

Q3. A loan of ₹25,220 is to be paid back in three equal annual installments. If the interest is compounded annually at 5% p.a., find the value of each installment.

- (A) ₹9,261
- (B) ₹9,000
- (C) ₹9,450
- (D) ₹8,820

SSC CPO 2023

Q4. A sum of ₹18,000 is lent at 20% p.a. compound interest, compounded half-yearly. What is the compound interest for the period of 1.5 years?

- (A) ₹6,242
- (B) ₹6,000
- (C) ₹5,760
- (D) ₹5,958

SSC CHSL Tier-II 2022

Q5. If the simple interest on a sum of money for 2 years at 5% p.a. is ₹4,000, what will be the compound interest on the same sum for the same period at the same rate?

- (A) ₹4,005
- (B) ₹4,200
- (C) ₹4,100
- (D) ₹4,050

SSC CGL Tier-II 2021

Q6. A sum of ₹15,000 is invested partly at 12% p.a. and the remaining at 10% p.a. simple interest. If the total interest at the end of 2 years is ₹3,344, how much money was invested at 10% p.a.?

- (A) ₹8,600
- (B) ₹8,000
- (C) ₹6,400
- (D) ₹7,500

SSC Selection Post Phase XI

Q7. A certain sum amounts to ₹4,600 after 3 years and ₹6,000 after 5 years at simple interest. Find the rate of interest per annum.

- (A) 28%
- (B) 20%
- (C) 25%
- (D) 30%

SSC MTS 2022

Q8. The compound interest on ₹30,000 at 7% p.a. is ₹4,347. Find the time period if the interest is compounded annually.

- (A) 2.5 years
- (B) 2 years
- (C) 3 years
- (D) 1.5 years

SSC CGL Tier-II 2022

Q9. A sum of money triples itself in 8 years at simple interest. In how many years will it become 8 times itself at the same rate?

- (A) 32 years
- (B) 21 years
- (C) 28 years
- (D) 24 years

SSC CPO 2022

Q10. At what rate per annum will ₹48,000 yield a compound interest of ₹7,566 in 2 years, compounded annually?

- (A) 8.5%
- (B) 7.5%
- (C) 7%
- (D) 8%

SSC CHSL Tier-II 2023

Q11. The difference between the compound interest compounded 8-monthly and the simple interest on a sum of ₹15,000 for 2 years at 12% p.a. is:

- (A) ₹144.40
- (B) ₹152.00
- (C) ₹149.76
- (D) ₹135.50

SSC CGL 2022

Q12. A person borrows ₹100,000 at 10% p.a. compound interest. He repays ₹30,000 at the end of the first year and ₹40,000 at the end of the second year. How much must he pay at the end of the third year to clear the debt?

- (A) ₹52,000
- (B) ₹50,000
- (C) ₹48,500
- (D) ₹51,700

SSC CGL Tier-II 2023

Q13. A sum is divided into three parts in the ratio 2:3:4. The parts are invested at SI at 4%, 5%, and 6% p.a. respectively. If the total SI earned in 3 years is ₹2,835, find the total sum invested.

- (A) ₹22,500
- (B) ₹18,000
- (C) ₹25,000
- (D) ₹20,250

SSC CPO 2021

Q14. If the rate of interest is 10% for the first year, 12% for the second year, and 15% for the third year, what is the compound interest on ₹25,000 for 3 years?

- (A) ₹11,000
- (B) ₹10,500
- (C) ₹9,800
- (D) ₹10,420

SSC CHSL 2021

Q15. The compound interest on a sum for the 3rd year is ₹1,452 and for the 4th year is ₹1,597.20. Find the principal sum.

- (A) ₹11,500
- (B) ₹10,000
- (C) ₹12,000
- (D) ₹15,000

SSC Selection Post Phase X

Q16. A man buys a scooter on cash down payment of ₹16,224 and promises to pay two more equal annual installments of ₹16,224 each. If the rate of interest is 4% p.a. compounded annually, what is the cash value of the scooter?

- (A) ₹48,000
- (B) ₹45,500
- (C) ₹46,800
- (D) ₹47,000

SSC CGL Tier-II 2023

Q17. A sum of money amounts to ₹10,080 in 3.5 years and ₹11,616 in 5.5 years at simple interest. Find the simple interest on the same sum for 4.5 years at double the rate.

- (A) ₹6,500
- (B) ₹6,912
- (C) ₹6,800
- (D) ₹7,200

SSC CGL 2020

Q18. A sum invested at compound interest amounts to ₹8,000 in 2 years and ₹27,000 in 5 years. Find the rate of interest per annum.

- (A) 45%
- (B) 50%
- (C) 60%
- (D) 40%

SSC CHSL Tier-II 2022

Q19. A certain sum amounts to ₹29,160 in 2 years and to ₹31,492.80 in 3 years on compound interest compounded annually. How much will be the simple interest on the same sum for 4 years at the same rate?

- (A) ₹8,400
- (B) ₹7,500
- (C) ₹9,000
- (D) ₹8,000

SSC CPO 2022

Q20. What is the compound interest on ₹12,000 for 1 year 3 months at 8% p.a. compounded annually?

- (A) ₹1,224
- (B) ₹1,150
- (C) ₹1,250
- (D) ₹1,200

SSC MTS 2021

Q21. A man lent out ₹9,000 at 6% p.a. and ₹12,000 at 8% p.a. simple interest. In how many years will the total interest earned be ₹7,500?

- (A) 5 years
- (B) 6 years
- (C) 4.5 years
- (D) 4 years

SSC CGL Tier-II 2021

Q22. If the difference between CI and SI on a sum for 2 years at 15% p.a. is ₹180, what is the simple interest on the same sum for 3 years at 10% p.a.?

- (A) ₹2,500
- (B) ₹2,400
- (C) ₹2,000
- (D) ₹3,000

SSC MTS 2022

Q23. A sum of ₹10,000 is invested in a scheme that offers 5% p.a. CI compounded half-yearly for the first year and 10% p.a. CI compounded annually for the second year. Find the total amount after 2 years.

- (A) ₹11,600.50
- (B) ₹11,550.00
- (C) ₹11,556.87
- (D) ₹11,480.00

SSC CGL 2022

Q24. Find the difference between the compound interest compounded yearly and half-yearly on ₹10,000 for 18 months at 10% p.a.

- (A) ₹25.00
- (B) ₹24.00
- (C) ₹26.25
- (D) ₹27.50

SSC Selection Post Phase XI

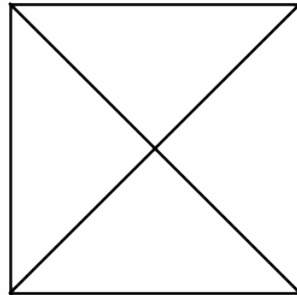
Q25. The simple interest on a sum of money is $\frac{9}{16}$ of the principal. If the number representing the rate percent is equal to the number of years, find the rate of interest.

- (A) 8%
- (B) 6.5%
- (C) 6%
- (D) 7.5%

SSC CHSL 2023

PART - B: REASONING (COUNTING OF FIGURES)

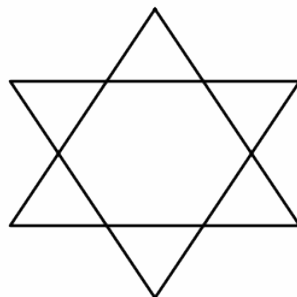
Q26. How many triangles are there in the given figure?



- (A) 6
- (B) 8
- (C) 10
- (D) 4

SSC CGL Tier-II 2021

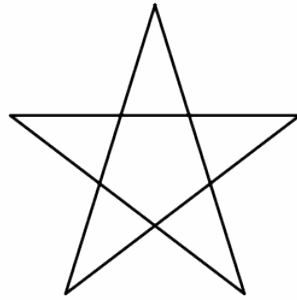
Q27. How many triangles are there in the given figure?



- (A) 12
- (B) 8
- (C) 10
- (D) 6

SSC CGL Tier-II 2022

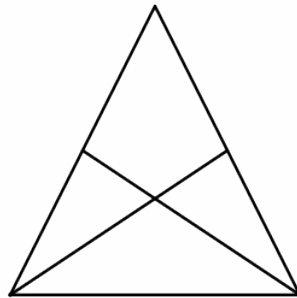
Q28. How many triangles are there in the given pentagram?



- (A) 12
- (B) 8
- (C) 10
- (D) 6

SSC CGL Tier-II 2021

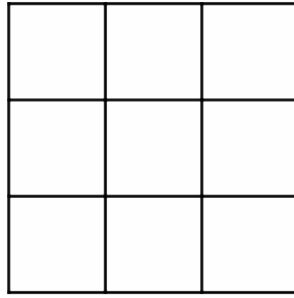
Q29. How many triangles are there in the given figure?



- (A) 6
- (B) 8
- (C) 10
- (D) 5

SSC CGL Tier-II 2022

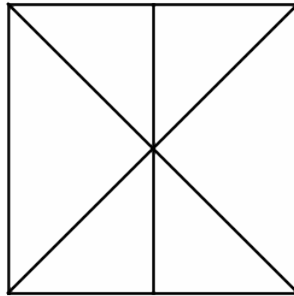
Q30. How many squares are there in the given grid?



- (A) 9
- (B) 14
- (C) 10
- (D) 16

SSC CGL Tier-II 2021

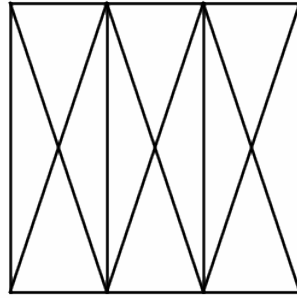
Q31. How many triangles are there in the given figure?



- (A) 18
- (B) 20
- (C) 16
- (D) 12

SSC CGL Tier-II 2022

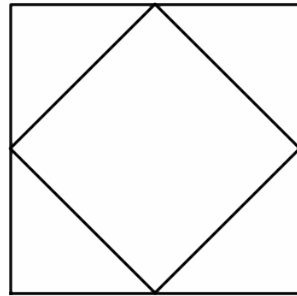
Q32. How many straight lines are required to draw the given figure?



- (A) 12
- (B) 14
- (C) 10
- (D) 16

SSC CGL Tier-II 2023

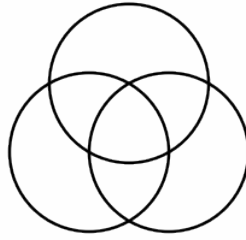
Q33. How many squares are there in the given figure?



- (A) 6
- (B) 7
- (C) 5
- (D) 4

SSC CGL Tier-II 2022

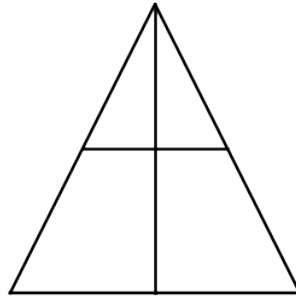
Q34. How many total enclosed bounded regions (simple areas) are formed?



- (A) 8
- (B) 5
- (C) 7
- (D) 6

SSC CGL Tier-II 2022

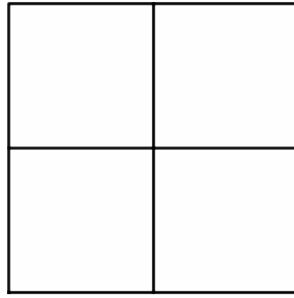
Q35. How many triangles are there in the given figure?



- (A) 6
- (B) 5
- (C) 7
- (D) 8

SSC CGL Tier-II 2021

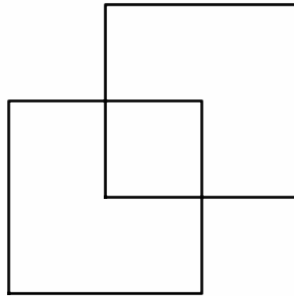
Q36. How many rectangles (including squares) are there?



- (A) 8
- (B) 9
- (C) 7
- (D) 10

SSC CGL Tier-II 2022

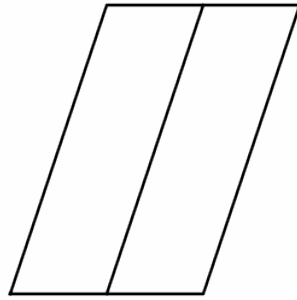
Q37. How many squares are there in the given figure?



- (A) 3
- (B) 4
- (C) 2
- (D) 5

SSC CGL Tier-II 2021

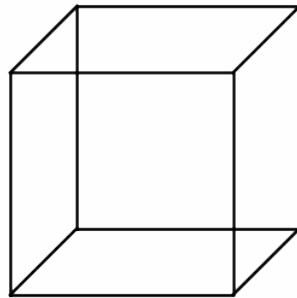
Q38. How many parallelograms are there in the given figure?



- (A) 3
- (B) 4
- (C) 5
- (D) 2

SSC CGL Tier-II 2023

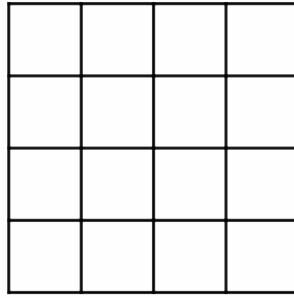
Q39. How many straight lines are required to construct this cube?



- (A) 12
- (B) 14
- (C) 10
- (D) 16

SSC CGL Tier-II 2022

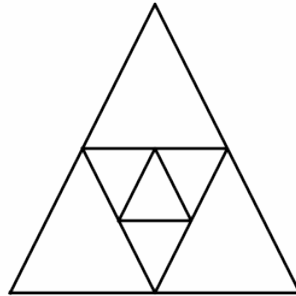
Q40. How many squares are there in the given 4x4 grid?



- (A) 16
- (B) 35
- (C) 30
- (D) 25

SSC CGL Tier-II 2023

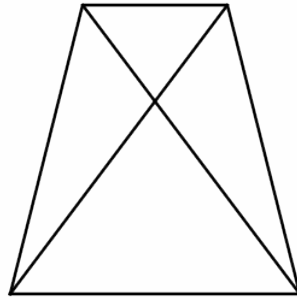
Q41. How many triangles are present in the given figure?



- (A) 9
- (B) 8
- (C) 11
- (D) 10

SSC CGL Tier-II 2022

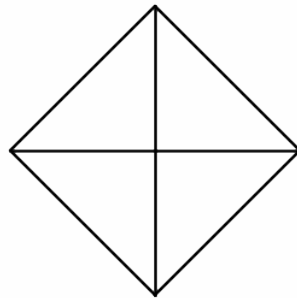
Q42. How many triangles are there in the given figure?



- (A) 6
- (B) 4
- (C) 10
- (D) 8

SSC CGL Tier-II 2023

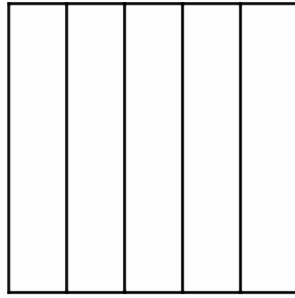
Q43. How many right-angled triangles are there?



- (A) 6
- (B) 10
- (C) 8
- (D) 4

SSC CGL Tier-II 2021

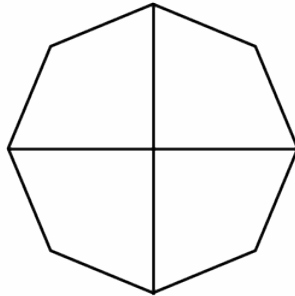
Q44. How many rectangles are there in the given figure?



- (A) 15
- (B) 5
- (C) 10
- (D) 20

SSC CGL Tier-II 2023

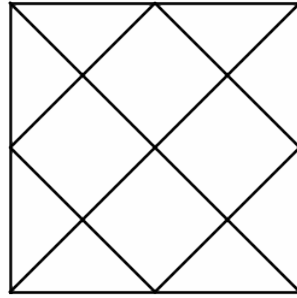
Q45. How many straight lines are there in the figure?



- (A) 10
- (B) 14
- (C) 12
- (D) 8

SSC CGL Tier-II 2021

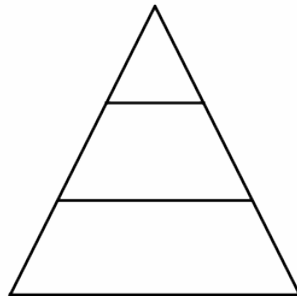
Q46. How many triangles are there in the given figure?



- (A) 16
- (B) 20
- (C) 24
- (D) 12

SSC CGL Tier-II 2022

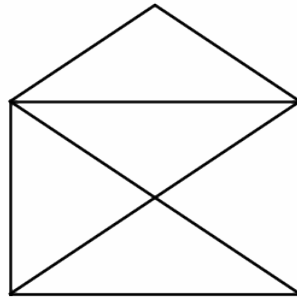
Q47. How many triangles are there in the given figure?



- (A) 4
- (B) 3
- (C) 6
- (D) 5

SSC CGL Tier-II 2023

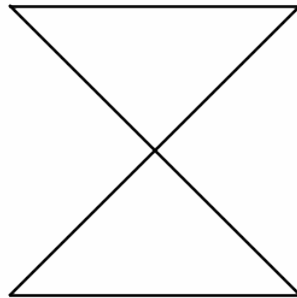
Q48. How many triangles are there in the given figure?



- (A) 9
- (B) 11
- (C) 10
- (D) 8

SSC CGL Tier-II 2021

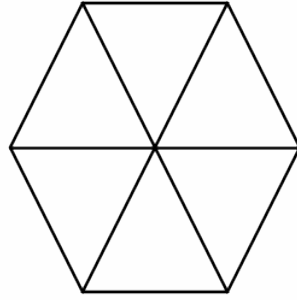
Q49. How many triangles are there in the given figure?



- (A) 10
- (B) 6
- (C) 8
- (D) 4

SSC CGL Tier-II 2022

Q50. How many triangles are there in the given figure?



- (A) 16
- (B) 18
- (C) 12
- (D) 14

SSC CGL Tier-II 2023